SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Industry Problem Task (5 QUESTIONS)

Course Code:	MLA0407	**Course Title:**	**Deep Learning**
CO Assessed:	CO4	**Assessment:**	Assessment 2: Industry Problem Task
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.***		
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Section / Batch:	AIML(2024)	**Date:**	29.08.2026

Code of Conduct
I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: CH. Durga Prasad

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Industry Problem Task

1. Industrial Defect Detection
A manufacturing company wants to automatically identify defective products from conveyor belt images. Design a transfer learning-based defect detection/classification solution using DenseNet or another suitable pre-trained CNN. Explain your model selection, feature extraction, fine-tuning strategy, and evaluation metrics.

2. Medical Diagnosis Support System
A hospital wants an AI system to assist doctors in identifying diseases from medical images. Develop a deep learning solution using transfer learning for medical image classification. Compare feature extraction and fine-tuning approaches and justify the final model.

3. Customer Review Sentiment Analysis
An e-commerce company receives thousands of customer reviews every day and wants to automatically understand customer sentiment. Design an RNN/LSTM/BiRNN-based sentiment analysis system to classify reviews as positive, negative, or neutral.

4. Predictive Text for a Mobile Application
A mobile keyboard company wants to provide intelligent next-word suggestions. Develop an LSTMbased next-word prediction model and explain how sequential data is processed using BPTT and how LSTM handles long-term dependencies.

5. Automated Customer Support
A multinational company wants to automatically translate customer queries between languages. Design an Encoder–Decoder-based Sequence-to-Sequence model for machine translation. Explain the role of the encoder, decoder, and sequence representation.

6. Social Media Content Monitoring
A social media platform wants to automatically classify posts based on sentiment or category. Develop and compare RNN, BiRNN, and LSTM models for the problem. Analyze which architecture is most suitable for understanding contextual information.

7. Agriculture Crop Disease Detection
An agricultural technology company wants to identify crop diseases from leaf images submitted by farmers. Develop a transfer learning-based image classification system using DenseNet. Apply data augmentation, feature extraction, and fine-tuning to improve performance.

8. Financial News Analysis
A financial organization wants to analyze news headlines and predict whether the market sentiment is positive, negative, or neutral. Design an LSTM or BiRNN-based sequential learning model and compare its performance with a basic RNN.

9. Automated Document Processing
A company receives a large number of scanned documents and wants to automatically classify them into categories such as invoices, resumes, reports, and contracts. Design a deep learning-based document classification system using transfer learning. Discuss preprocessing, feature extraction, model development, and fine-tuning.

10. Smart Traffic Monitoring
A smart city project wants to classify images captured from traffic cameras into categories such as cars, buses, motorcycles, and emergency vehicles. Develop a transfer learning-based traffic classification model using a pre-trained DenseNet architecture and optimize it for real-time deployment.

1. SOCIAL MEDIA CONTENT MONITORING

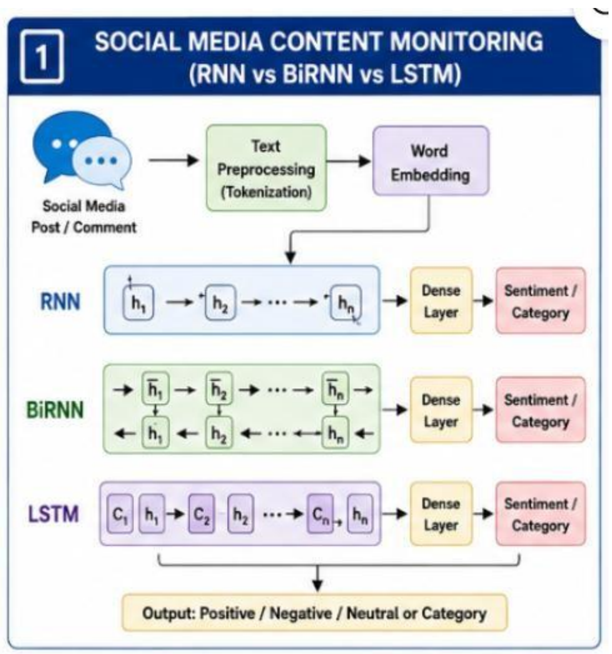
Problem

A social media platform wants to automatically classify posts based on sentiment or category using RNN, BiRNN and LSTM.

Answer

The text is first converted into numerical sequences using tokenization and word embeddings. Then different sequential models are trained and compared.

Architecture



RNN

Processes text from left to right and remembers previous words.

BiRNN

Processes text in both directions, so it uses past and future context.

LSTM

Uses memory cells and gates to remember important information for longer sequences.

Comparison

Model	**Context**	**Long-Term Memory**
RNN	Past	Low
BiRNN	Past + Future	Medium
LSTM	Sequential context	High

Conclusion

LSTM or BiRNN is generally more suitable than basic RNN for understanding contextual information. LSTM is especially useful for long posts because it handles long-term dependencies better.

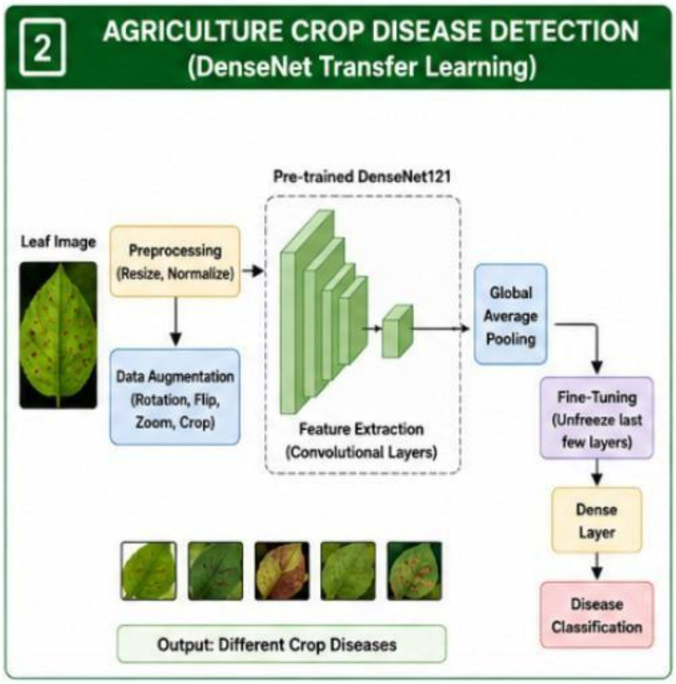
2. AGRICULTURE CROP DISEASE DETECTION

Problem Identify crop diseases from leaf images using transfer learning with DenseNet.

Answer

A pre-trained **DenseNet** can be used to extract important visual features from crop leaf images.

Architecture



Preprocessing

- Resize images
- Normalize pixel values
- Remove corrupted images
- Split into training, validation and testing sets **Data Augmentation** Apply:
- Rotation
- Flipping
- Zooming
- Cropping

This increases data variety and reduces overfitting.

Feature Extraction

Initially, DenseNet layers are frozen and used to extract features such as:

Edges → Textures → Leaf Patterns → Disease Features

Fine-Tuning

The last few DenseNet layers are unfrozen and trained with a small learning rate so that the model adapts to crop disease patterns.

Evaluation

Use:

- Accuracy

- Precision
- Recall
- F1-score
- Confusion Matrix

Conclusion

DenseNet with augmentation, feature extraction and fine-tuning can provide accurate crop disease detection while reducing training time.

3. FINANCIAL NEWS ANALYSIS

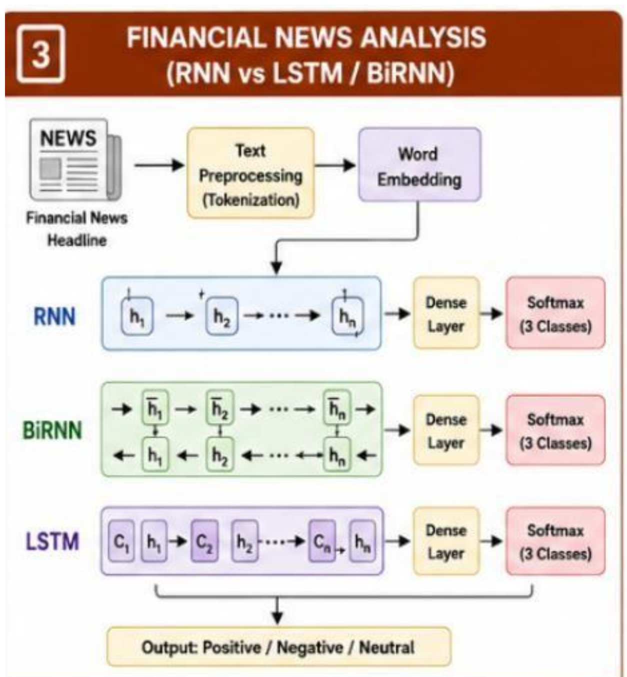
Problem

Predict whether financial news sentiment is positive, negative or neutral using LSTM/BiRNN and compare it with a basic RNN.

Answer

Financial headlines are sequential text, so an LSTM or BiRNN can learn relationships between words and predict market sentiment.

Architecture



Basic RNN

Text → Embedding → RNN → Dense → Softmax

RNN can capture short-term dependencies but may struggle with long-term relationships.

LSTM

LSTM uses gates to control information:

Input → LSTM → Memory → Dense → Sentiment

It can retain important information for longer sequences.

Comparison

RNN	LSTM/BiRNN
Simple architecture	More advanced
Short-term context	Better contextual learning
Faster	More computationally expensive
Vanishing-gradient problem	LSTM reduces this problem

Conclusion

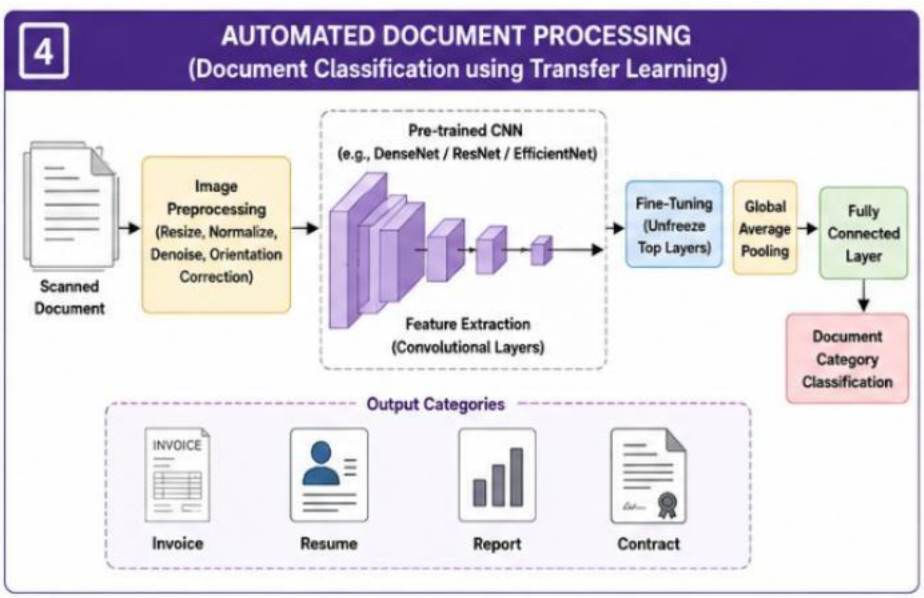
LSTM/BiRNN is preferred because financial headlines may contain important relationships between words that are separated in the sentence.

4. AUTOMATED DOCUMENT PROCESSING

Problem Classify scanned documents into invoices, resumes, reports and contracts using transfer learning. Answer

Transfer learning can use a pre-trained CNN to extract visual features from scanned documents.

Architecture



Preprocessing

- Convert scanned pages into images
- Resize images
- Normalize pixel values
- Remove noise
- Correct image orientation

Feature Extraction

The pre-trained CNN learns:

Edges → Text Layout → Tables → Document Patterns

Model Development

1. Load a pre-trained CNN.
2. Remove the original classifier.
3. Add new classification layers.
4. Freeze the base layers.
5. Train the new classifier.
6. Unfreeze selected layers.
7. Fine-tune using a low learning rate.

Output

Document
↓
↓ ↓ ↓ ↓
Invoice Resume Report Contract

Evaluation Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Conclusion

Transfer learning reduces training requirements and allows the system to automatically classify different document types based on their visual and structural features.

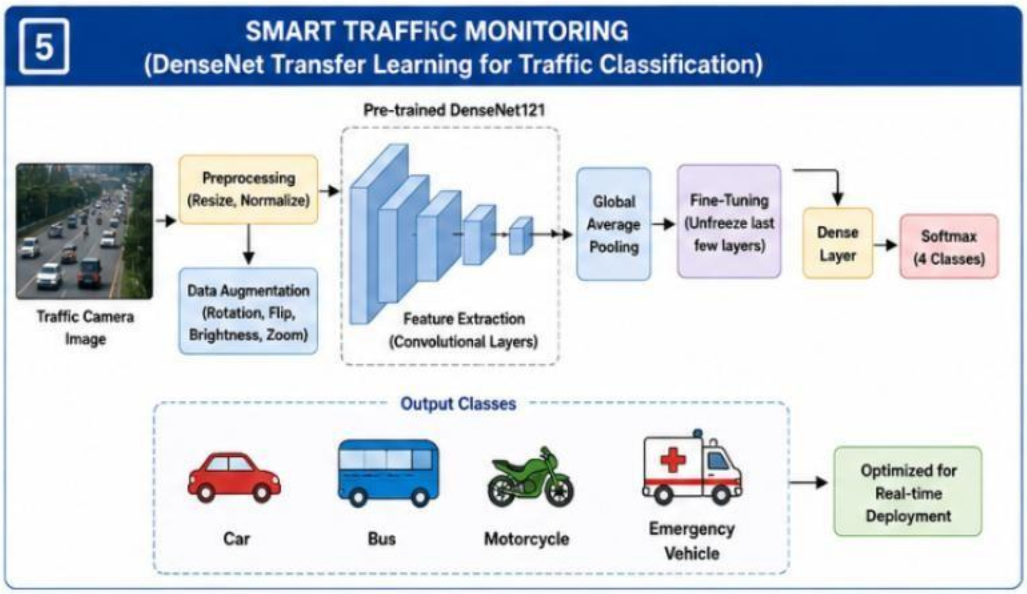
5. SMART TRAFFIC MONITORING

Problem

Classify traffic-camera images into cars, buses, motorcycles and emergency vehicles using a pre-trained DenseNet and optimize it for real-time deployment. Answer

A pre-trained **DenseNet** can be used to extract vehicle features and classify traffic images.

Architecture



Classes

Image
↓
↓ ↓ ↓
Car Bus Motorcycle
↓
Emergency Vehicle

Training

1. Collect traffic images.
2. Resize and normalize images.
3. Apply augmentation.
4. Load pre-trained DenseNet.
5. Freeze initial layers.
6. Train the classification layer.
7. Fine-tune deeper layers.
8. Evaluate the model.

Real-Time Optimization

To achieve faster prediction:

- Use a smaller input image size.
- Use batch processing where appropriate.
- Remove unnecessary layers.
- Use model quantization.
- Use GPU/edge hardware.
- Optimize the trained model for inference. **Evaluation** Use:

- Accuracy
- Precision
- Recall
- F1-score
- Inference time

For real-time systems, **inference speed is very important**.

Conclusion

DenseNet transfer learning provides strong feature extraction, while fine-tuning improves traffic-specific classification. Model optimization helps achieve accurate and fast real-time vehicle detection.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________ Date: ____________________	Signature: ________________ Date: ____________________	Signature: ________________ Date: ____________________